

Kylee Friederichs

Email: kylee.friederichs@tamu.edu

Phone: 949-751-7868

Location: College Station, TX

EDUCATION:

Doctor of Philosophy in Computer Engineering

Texas A&M University, 01/2025 - Present

GPA: 4.0/4.0

Master of Engineering in Computer Engineering

Texas A&M University, 01/2023 - 12/2024

GPA: 4.0/4.0

Bachelor of Arts in Computer Science

Minor in Space

University of Colorado Boulder, 08/2018 - 12/2022

GPA: 3.855/4.0

RESEARCH EXPERIENCE:

VirtuHS

Texas A&M University, 08/2024 - Present

- Developing VirtuHS, a framework to integrate virtual reality (VR) collaborative learning applications into high school education by focusing on key design guidelines for effective interactions.

Ocean of Memories

Texas A&M University, 11/2023 - 10/2024

- Collaborated with artists-in-residence Chromic Duo to design an AR image tracking and content management system in Unity for an immersive sound walk experience based on UN's SDG-14: Life Below Water, integrating GPS-based activation and diverse AR content.

American Bureau of Shipping (ABS) Hazard Perception Analysis Project

Texas A&M University, 12/2023 - 08/2024

- Contributed to assessing the feasibility and safety of augmented reality (AR) for maritime surveyors by constructing a mock ABS ship room with sensors and running hazard perception tests.

Increase Visitation in National Parks - AR Markers

Texas A&M University, 09/2023 - 08/2024

- Researched the impact of environmental factors on AR marker tracking in Costa Rica's National Parks, optimizing materials and scaling for outdoor performance while analyzing how marker material changes affect image tracking accuracy.

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NOTABLE PROJECTS:

Virtual Reality Project Series

Texas A&M University, 01/2024 - 05/2024

- Celestial Putt Putt
 - Space-themed VR mini-golf game with three different courses.
- Labyrinth - Quest of the Idol
 - Players must recover a lost idol and escape a maze before being captured.
- InfinityExpo
 - Infinitely expanding VR museum populated with images and text from Wikipedia.

Dynamic Subdivision Visualizer

Texas A&M University, 10/2023 - 12/2023

- Augmented an interactive 3D visualization tool in Three.js for subdivision surfaces, enabling .OBJ model uploads, loop subdivision, and a color-coded scheme for enhanced user experience.

PUBLICATIONS AND CONFERENCES:

- Ocean of Memories: Raising Marine Awareness through Creativity and Augmented Reality
International Symposium on Electronic Art - 05/2025
- Analyzing the Impact of Augmented Reality Head-Mounted Displays on Workers' Safety and Situational Awareness in Hazardous Industrial Settings
arXiv - 03/2025

MEDIA APPEARANCES:

- The Versatelist Podcast
Immersive Learning Research Network 12/2025
- Artists Bring Augmented Reality To Aggie Park Through 'Ocean of Memories' Soundwalk
College of Performance, Visualization & Fine Arts - Texas A&M University, 12/2024
- Ocean Of Memories' Soundwalk Collaboration With Chromic Duo Brings Augmented Reality Experience To Aggie Park
College of Performance, Visualization & Fine Arts - Texas A&M University, 09/2024
- 13th Annual Faculty Submitted Student Work Exhibit
SIGGRAPH - 07/2024
- Safety In Maritime Surveying: Dr. Edgar Rojas-Muñoz Leading Research With Augmented Reality Devices
College of Performance, Visualization & Fine Arts - Texas A&M University, 06/2024